

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

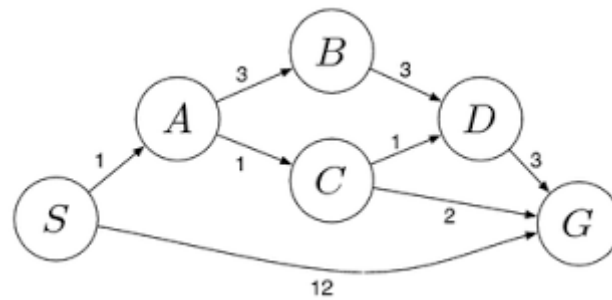
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I P POORNA CHANDU(192412199)(Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

QUESTION 1:

1. State Space

A state is represented as (x, y) , where:

- x = amount of water in the 4-gallon jug (0–4)
- y = amount of water in the 3-gallon jug (0–3)

Example states: $(0,0)$, $(4,0)$, $(1,3)$, $(2,0)$.

2. Initial State

$(0,0)(0,0)(0,0)$

Both jugs are empty.

3. Goal State

$(2,0)(2,0)(2,0)$

The 4-gallon jug contains exactly 2 gallons.

4. Operators (Actions)

- Fill the 4-gallon jug.
- Fill the 3-gallon jug.
- Empty the 4-gallon jug.
- Empty the 3-gallon jug.
- Pour water from the 4-gallon jug to the 3-gallon jug.
- Pour water from the 3-gallon jug to the 4-gallon jug.

5. Solution Path

Step	State (4-gallon, 3-gallon)	Action
1	$(0,0)$	Initial state
2	$(0,3)$	Fill the 3-gallon jug
3	$(3,0)$	Pour 3-gallon jug into 4-gallon jug
4	$(3,3)$	Fill the 3-gallon jug again
5	$(4,2)$	Pour into the 4-gallon jug until it is full
6	$(0,2)$	Empty the 4-gallon jug
7	$(2,0)$	Pour the remaining 2 gallons into the 4-gallon jug

Final Answer The goal state $(2,0)$ is reached. Hence, the 4-gallon jug contains exactly 2 gallons of water, which satisfies the problem requirement.

QUESTION 2:

i. Percepts (What the agent perceives)

The Mars Rover receives information from its sensors, such as:

- Images from cameras.
- Rock and soil characteristics.
- Temperature and atmospheric conditions.
- Terrain information (rocks, slopes, sand).
- Position and navigation data.
- Battery level and system health.

ii. Operating Environment

The operating environment of the Mars Rover is:

- Partially Observable: The rover cannot observe the entire environment at once.
- Deterministic: Actions usually produce predictable results.
- Sequential: Current actions affect future actions.
- Dynamic: Environmental conditions such as dust storms may change.
- Continuous: Time, movement, and location change continuously.
- Single Agent: The rover performs tasks independently.

iii. Actions (What the agent can do)

The Mars Rover can:

- Move forward, backward, left, or right.
- Capture images and videos.
- Collect rock and soil samples.
- Analyze samples using onboard instruments.
- Avoid obstacles.
- Transmit collected data to Earth.
- Recharge using solar panels (when possible).

iv. Performance Measure (How the agent is evaluated)

The performance of the Mars Rover is evaluated based on:

- Accuracy of collected scientific data.
- Number of successful sample collections.
- Safe navigation without collisions.
- Efficient use of battery power.
- Successful communication with Earth.

v. Suitable Agent Architecture

The most suitable architecture is a Utility-Based Agent.

Reason:

- The Mars Rover must choose the best action among several alternatives.
- It considers factors such as safety, battery power, terrain difficulty, and scientific value before making decisions.

QUESTION 3:

Initial State:

Empty 8×8 chessboard.

Goal State:

Place 8 queens so that no two queens are in the same row, column, or diagonal.

Operators:

- Place a queen in a safe position.
- Move to the next row.
- Backtrack if a conflict occurs.

Search Strategy:

Backtracking (Depth-First Search)

Solution Steps:

1. Place Queen 1 at **(Row 1, Column 1)**.
2. Place Queen 2 at **(Row 2, Column 5)**.
3. Place Queen 3 at **(Row 3, Column 8)**.
4. Place Queen 4 at **(Row 4, Column 6)**.
5. Place Queen 5 at **(Row 5, Column 3)**.
6. Place Queen 6 at **(Row 6, Column 7)**.
7. Place Queen 7 at **(Row 7, Column 2)**.
8. Place Queen 8 at **(Row 8, Column 4)**.

Final Solution

Queen	Position
Q1	1,1
Q2	2,5
Q3	3,8
Q4	4,6
Q5	5,3
Q6	6,7
Q7	7,2
Q8	8,4

Result:

All 8 queens are placed successfully, and no two queens attack each other.

QUESTION 4:

Goal-Based Agent

Reason:

The customer's goal is to travel from the source location to the destination. The agent selects the most suitable cab (Mini, Micro, Sedan, Shared, Prime, etc.) based on the customer's preference and availability.

CODE:

```
print("===== OLA CAB BOOKING =====")
```

```
print("Enter Source Location: Chennai")
```

```
print("Enter Destination: Tambaram")
```

```
print("Select Cab Type:")
```

```
print("1. Mini")
```

```
print("2. Micro")
```

```
print("3. Sedan")
```

```
print("4. Shared")
```

```
print("5. Prime")
```

```
choice = int(input("Enter your choice: "))
```

```
if choice == 1:
```

```
    fare = 250
```

```
    print("\nCab Type : Mini")
```

```
elif choice == 2:
```

```
    fare = 180
```

```
    print("\nCab Type : Micro")
```

```
elif choice == 3:
```

```
    fare = 350
```

```
    print("\nCab Type : Sedan")
```

```
elif choice == 4:
```

```
    fare = 150
```

```
    print("\nCab Type : Shared")
```

```
elif choice == 5:
```

```
fare = 500

print("\nCab Type : Prime")
else:

    print("\nInvalid Cab Selection!")

    exit()

print("Checking Cab Availability...")
print("Cab Available!")
print("Booking Confirmed.")
print("Driver Assigned Successfully.")
print("Estimated Fare = Rs.", fare)
print("Happy Journey!")
```

OUTCOME:

===== OLA CAB BOOKING =====

Enter Source Location: Chennai

Enter Destination: Tambaram

Select Cab Type:

1. Mini
2. Micro
3. Sedan
4. Shared
5. Prime

Enter your choice: 3

Cab Type : Sedan

Checking Cab Availability...

Cab Available!

Booking Confirmed.

Driver Assigned Successfully.

Estimated Fare = Rs. 350.00 ,Happy Journey!

QUESTION 5:

Initial State

Start node = S

Goal State

Destination node = G

Step 1: Expand S

Possible paths from S:

Path	Cost
$S \rightarrow A$	1
$S \rightarrow G$	12

Choose the least-cost path: $S \rightarrow A$ (1)

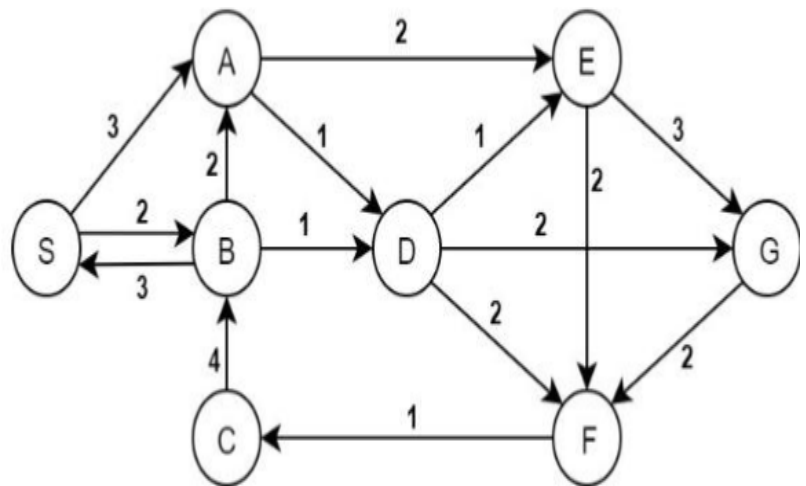
Step 2: Expand A

From A:

Path	Cost
$S \rightarrow A \rightarrow B$	$1 + 3 = 4$
$S \rightarrow A \rightarrow C$	$1 + 1 = 2$
$S \rightarrow G$	12

Choose the least-cost path: $S \rightarrow A \rightarrow C$ (2)

Step 3: Expand C



From C:

Path	Cost
$S \rightarrow A \rightarrow B$	4
$S \rightarrow A \rightarrow C \rightarrow D$	$2 + 1 = 3$
$S \rightarrow A \rightarrow C \rightarrow G$	$2 + 2 = 4$
$S \rightarrow G$	12

Choose the least-cost path: $S \rightarrow A \rightarrow C \rightarrow D$ (3)

Step 4: Expand D

From D:

Path	Cost
$S \rightarrow A \rightarrow B$	4
$S \rightarrow A \rightarrow C \rightarrow G$	4
$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	$3 + 3 = 6$
$S \rightarrow G$	12

The smallest cost among goal paths is 4.

UCS Search Order

- S (0)
- A (1)
- C (2)

- D (3)
- G (4)

Least-Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost = 4